Project Proposal: AI Technical Instruction to Image Generator

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Motivation:

Our goal is to generate step-by-step images to accompany difficult, written technical instructions. Our project aims to help people who may need more visual assistance as guidance for learning and completing technical tasks such as origami, tying knots, making recipes, building Legos, etc.

Description:

We propose an agentic workflow consisting of a host orchestrator agent and multiple remote image generation agents. The host orchestrator will process raw technical instructions, simplify them, and divide them into discrete subtasks for distribution to the remote agents. These agents will generate images to accompany their assigned tasks. The workflow will be designed to generate a variable number of images corresponding to the subtasks identified by the host orchestrator. We will compare this workflow against a single-agent baseline that generates images directly from the same raw instructions.

Plan of Execution:

To implement the system, we are investigating A2A through the Beeai Framework with locally hosted Ollama models. We are also investigating ComfyUI. We will evaluate both systems on tasks ranging from simple to complex, including drawing, Lego building, baking, and origami folds. For our performance metric, we will use human feedback to compare the usefulness of the generated images for interpreting and completing the

instructions. Ideally, we would poll the class to determine whether the single-agent baseline or agentic workflow provides more useful visual guidance.